



PHILIPPINE STATE COLLEGE OF AERONAUTICS



INSTITUTE OF COMPUTER STUDIES

Name of Institute



BS INFORMATION TECHNOLOGY

*with specialization in
Aviation Information Technology
Name of Program*



CAPSTONE PROJECT 2

PRELIM MODULES



by:

ASSOC. PROF. MARY ANN F. ABALLIAR-VISTA, MEAM



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COURSE OVERVIEW

Course Information

Welcome to **IT411: Aviation Capstone Project 2**. This chapter introduces the essential information that every student should know before beginning the course. It provides an overview of the course description, intended learning outcomes, program learning outcomes, learning outcomes alignment, course requirements, grading system, course policies, academic integrity, AI-use disclosure, and recommended references. Familiarity with these course components will help students understand the expectations, responsibilities, and standards required for the successful completion of their capstone project.

Course Code: IT411

Course Title: Aviation Capstone Project 2

Course Type and Credit: Lecture (3 Units), 54 Hours (Project-Based / Research)

Pre-requisite: IIT327

Co-requisite: None

Aviation Capstone Project 2 is the continuation and culminating implementation phase of Aviation Capstone Project 1. It focuses on the completion, validation, evaluation, defense, revision, deployment, and archiving of an approved information technology-based capstone system designed for aviation, airline, institutional, or industry-related applications.

The course covers project orientation and readiness review; presentation of proposal-defense outputs; development of pre-assessment, ISO/IEC 25010, and Technology Acceptance Model (TAM)-based evaluation instruments; validation procedures; reliability testing; preparation of Chapters IV and V of the research manuscript; mock final defense; final defense; revision presentation; deployment; submission of electronic and printed project outputs; and research clearance.

As the terminal course in the BS Aviation Information Technology program, Aviation Capstone Project 2 integrates research methodology, software and system development, quality evaluation, statistical analysis, technical writing, project management, ethical research practices, AI-tool disclosure, data privacy, intellectual property awareness, and professional collaboration to prepare students for industry and professional practice.

1.2 Course Learning Outcomes (CLO)

At the end of the semester, students are expected to:



Course Learning Outcomes (CLO)

CLO 1. Evaluate approved capstone proposals and proposal-defense outputs using adviser/panel findings, routing documents, and institutional capstone requirements to determine readiness for system completion and final documentation with at least **80% compliance** based on the capstone readiness rubric.

CLO 2. Develop, refine, and demonstrate IT-based capstone systems using approved system designs, development tools, and stakeholder requirements to address target beneficiary needs with at least **80% functionality** based on the system evaluation rubric.

CLO 3. Design, validate, and administer research and system-evaluation instruments using pre-assessment tools, ISO/IEC 25010 questionnaires, TAM-based questionnaires, validation forms, and reliability procedures to gather valid and reliable data from target beneficiaries, IT experts, and users with ethical compliance.

CLO 4. Analyze and interpret research and system-evaluation data using appropriate research design, data-gathering techniques, sampling procedures, and statistical treatment to prepare Chapter IV tables, results, and discussions accurately aligned with the research objectives.

CLO 5. Synthesize research findings by preparing Chapter V summary, conclusions, and recommendations based on validated results, system evaluation evidence, and project objectives with logical consistency and practical relevance.

CLO 6. Produce complete capstone documentation and technical deliverables using the prescribed NAAP/ICS capstone layout, citation requirements, formatting standards, appendices, revision logs, source files, and system files with institutional compliance.

CLO 7. Present and defend the capstone system and manuscript before advisers, panel members, users, and institutional evaluators using oral presentation, system demonstration, visual materials, and evidence-based responses with confidence, technical accuracy, and professionalism.

CLO 8. Demonstrate professionalism, teamwork, ethical research practice, proper AI-tool disclosure, data privacy awareness, and respect for intellectual property throughout system development, defense, deployment, and clearance processes.



1.3 Learning Outcomes Alignment

The Aviation Capstone Project 2 course follows the principles of Outcome-Based Education (OBE). Each learning activity is aligned with the Program Learning Outcomes (PLO), Course Learning Outcomes (CLO), Module Learning Outcomes (MLO), and Topic Learning Outcomes (TLO) to ensure that students progressively acquire the competencies required for successful completion of the capstone project.

The relationship among these learning outcomes is illustrated below.



This alignment ensures that every lesson, activity, consultation, presentation, and assessment contributes directly to the attainment of the intended course and program outcomes.

1.4 Course Requirements

To successfully complete the course, students are expected to accomplish all required academic, research, and technical deliverables according to the prescribed schedule.

Major course requirements include:

- Approved Capstone Project Proposal
- Revised Research Manuscript
- System Prototype
- Evaluation Instruments
- Validation Documents
- Consultation Records
- Research-Day Participation
- Mock Final Defense
- Final Defense
- Revision Presentation
- System Deployment
- Electronic Submission of Project Files
- Hardbound Research Manuscript
- Research Clearance

Students are expected to submit all requirements on or before the prescribed deadlines.

1.5 Grading System

The grading system follows the official institutional policies.

- The semestral grade is computed based on the average of the Preliminary, Midterm, and Final period grades.
- No grading component shall exceed **30%** of the total grade.
- Attendance is monitored but is not included in the computation of grades.
- Presentations, technical reports, manuscripts, system demonstrations, practical outputs, and other performance tasks shall be evaluated using approved rubrics.

- Computation of grades is 0-Based with a 75% passing rate
- **SEMESTER GRADE** = (Prelim Grade + Midterm Grade + Final Term Grade) / 3

CRITERIA	PERCENTAGE
Major Capstone Presentations / Examinations (Preliminary Chapter IV Presentation, Mock Defense, Final Defense, Final Submission)	30%
System Development, Deployment, and Technical Deliverables (System prototype, functionality, deployment evidence, source code, prototype/device if applicable)	25%
Research Documentation and Manuscript Outputs (Chapters IV-V, revised manuscript, final manuscript, appendices, references)	25%
Consultation, Validation, Revision, Peer/Adviser Evaluation, and Compliance Outputs (Validation forms, reliability evidence, consultation logs, revision matrix, routing and clearance documents)	20%
TOTAL	100%

1.6 Course Policies

1. Attendance and Participation. Attendance and active participation during orientation, consultation, research day, defense, deployment, and clearance sessions are essential. Students who fail to attend more than 20% of class meetings may be given a remark of F.A. or Failure due to Absences, subject to institutional policy.
2. Tardiness. A grace period of 15 minutes is allowed. Students who arrive beyond the grace period will be marked tardy. Two tardiness marks are equivalent to one absence.
3. Group Roles and Accountability. Each group must assign clear roles such as project leader, documentation lead, developer, researcher, statistician coordinator, and deployment coordinator. Individual accountability may be considered when evaluating non-participation or unequal contribution.



4. Consultation and Research Day. Groups are required to consult their capstone adviser and, when necessary, the statistician, validators, or panel members. Consultation logs, adviser comments, and revision matrices shall serve as evidence of progress.
5. Deadlines and Submission. All outputs must be submitted on or before the deadline in the required format. Late submissions will be subject to institutional policy and/or deductions announced by the instructor unless a valid reason is approved.
6. Research Instruments, Validity, and Reliability. Survey questionnaires shall be aligned with project objectives and must undergo appropriate validation. Reliability testing shall be conducted when applicable before or during data-gathering activities.
7. Academic Integrity and Honesty. Plagiarism, fabricated data, falsified validation forms, fabricated survey responses, false deployment evidence, and unauthorized replication of systems are strictly prohibited. Violations may result in zero credit and institutional disciplinary action.
8. Responsible Use of AI Tools. AI tools may be used for grammar checking, outlining, debugging support, and productivity assistance only when properly reviewed, revised, and disclosed. AI tools must not be used to fabricate data, sources, survey responses, system functionalities, or complete manuscript content.
9. Data Privacy and Ethical Practice. Students must observe confidentiality, informed consent, ethical data-gathering, responsible handling of personal information, and proper storage of research and system data.
10. Technology Use. Groups must maintain organized digital folders, version-controlled files, source code backups, project trackers, and deployment records. Technology shall be used responsibly and professionally.



11. Defense and Presentation. Groups must be prepared with final slides, system demo, manuscript, revision matrix, and supporting evidence. Professional conduct and proper communication are required during mock defense, final defense, re-defense, and revision presentation.
12. System Deployment and Intellectual Property. Deployment must be documented through reports, screenshots, user feedback, acceptance evidence, and credentials where applicable. Students must respect intellectual property rights, licenses, data ownership, confidentiality agreements, and institutional guidance on copyright, utility model, or patent protection where applicable.
13. Final Clearance and Completion. Final grading and clearance are subject to completion of revised documentation, deployed system or prototype/device when applicable, e-files, hardbound manuscript, approval sheets, research clearance, and required institutional archiving documents.

1.7 Academic Integrity and AI-Use Disclosure

Academic integrity is fundamental to the conduct of research. Students shall produce original work, properly acknowledge all sources, and avoid plagiarism, fabrication, falsification, and other forms of academic dishonesty.

The responsible use of Artificial Intelligence (AI) tools is permitted only when authorized by the instructor or adviser. Any use of AI-assisted technologies for writing, programming, data analysis, editing, or content generation must be properly disclosed in accordance with institutional policies. Students remain fully responsible for verifying the accuracy, originality, and ethical use of all submitted work.

1.8 Honesty and Safety Reminder

Honesty Clause

Students are expected to uphold academic integrity by submitting original work, properly acknowledging sources, and using Artificial Intelligence (AI) tools responsibly. Plagiarism, fabrication, unauthorized copying, and other forms of academic dishonesty are strictly prohibited.

Safety Clause



Students shall observe institutional policies, data privacy, information security, and safety protocols during all capstone activities, including consultations, system development, testing, deployment, and documentation.

1.9 Recommended References

Books

Garcia, L. C., et al. (2016). **Outcomes-Based Practical Guide to Thesis and Capstone Project Writing in Computing**. Mindshapers Co., Inc.

Howell, D. C. (2008). **Fundamental Statistics for the Behavioral Sciences** (6th ed.).

Institutional References

Philippine State College of Aeronautics. (2025). *IT411: Aviation Capstone Project 2 course syllabus* (Version 2, Revision 1, 1st Semester, S.Y. 2025–2026). Institute of Computer Studies.

Philippine State College of Aeronautics. (2013). *PhilSCA student research handbook: AY 2013–2016*.

Standards

International Organization for Standardization. (2023). **ISO/IEC 25010:2023: Systems and software engineering—Systems and software Quality Requirements and Evaluation (SQuaRE)—Product quality model**. <https://www.iso.org/standard/78176.html>

Davis, F. D. (1989). **Perceived usefulness, perceived ease of use, and user acceptance of information technology**. *MIS Quarterly*, 13(3), 319–340. <https://doi.org/10.2307/249008>

Davis, F. D., Bagozzi, R. P., & Warshaw, P. R. (1989). **User acceptance of computer technology: A comparison of two theoretical models**. *Management Science*, 35(8), 982–1003. <https://doi.org/10.1287/mnsc.35.8.982>





Online Resources

1. Commission on Higher Education (CHED): <https://ched.gov.ph>
2. IEEE Xplore Digital Library: <https://ieeexplore.ieee.org>
3. ACM Digital Library: <https://dl.acm.org>
4. Google Scholar: <https://scholar.google.com>





PRELIMINARY PERIOD

Module 1

Course Orientation Capstone Project Roadmap

Week: 1 | Time Allotment: 3 Hours

Learning Outcomes

Course Learning Outcomes (CLO)

This module contributes to the attainment of the following Course Learning Outcomes:

CLO 1. Evaluate approved capstone proposals and proposal-defense outputs using adviser/panel findings, routing documents, and institutional capstone requirements to determine readiness for system completion and final documentation with at least 80% compliance based on the capstone readiness rubric.

CLO 8. Demonstrate professionalism, teamwork, ethical research practice, proper AI-tool disclosure, data privacy awareness, and respect for intellectual property throughout system development, defense, deployment, and clearance processes.

Module Learning Outcome (MLO)

MLO 1: Analyze Capstone Project 2 requirements, schedules, group responsibilities, consultation protocols, ethics, AI-use disclosure, and final deliverables to establish readiness for the implementation phase.

Topic Learning Outcomes (TLO)

At the end of this module, the learners should be able to:

TLO 1. Explain the course requirements, grading system, timeline, consultation flow, defense expectations, and final deliverables.





TLO 2. Prepare a group role distribution, project tracker, and consultation plan aligned with institutional capstone requirements.

Learning Objectives

Upon successful completion of this module, the students will be able to:

1. Explain the overall objectives and expectations of Aviation Capstone Project 2.
2. Describe the grading system, consultation procedures, and semester timeline.
3. Identify the required academic, research, and technical deliverables for the course.
4. Recognize the importance of professionalism, ethical research practices, data privacy, intellectual property, and responsible AI use.
5. Organize the capstone group by assigning member roles and preparing a consultation and project implementation plan.

Time Commitment

ACTIVITY	ESTIMATED TIME
Reading the Module	45 minutes
Lesson Proper	60 minutes
Guided Learning Activities	45 minutes
Formative Assessment	30 minutes
Total	3 Hours

Required Resources

Students are encouraged to prepare the following learning resources before starting this module:

- Course Syllabus
- Learning Management System (LMS) Account





- Google Classroom
 - Google Drive
 - Capstone Manual
 - Project Tracker Template
 - AI-use Disclosure Guide
 - Notebook or Digital Note-taking Application
 - Laptop/Desktop Computer with Internet Access
-

Module Overview

The first module serves as the foundation of the Aviation Capstone Project 2 course. It introduces students to the course requirements, grading system, consultation procedures, project timeline, defense expectations, ethical responsibilities, AI-use disclosure, and the complete set of deliverables required throughout the semester.

Students will also establish their project organization by identifying individual roles, preparing a project tracker, and developing a consultation plan that will guide the successful implementation and completion of their capstone project. By the end of this module, learners are expected to understand the overall direction of the course and prepare for the succeeding modules with a clear understanding of institutional expectations and professional responsibilities.

Introduction

Welcome to **IT411: Aviation Capstone Project 2**, the culminating course of the Bachelor of Science in Aviation Information Technology (BSAIT) program. Congratulations on reaching this significant milestone in your academic journey. Your successful completion of Aviation Capstone Project 1 has equipped you with a research proposal, a system design, and a project direction. In this course, your responsibility is to transform these initial outputs into a fully developed, validated, evaluated, defended, and deployable information technology solution that addresses the needs of your identified beneficiaries.

Unlike previous courses that primarily focused on acquiring knowledge and technical skills, Aviation Capstone Project 2 emphasizes the practical application of those competencies in solving real-world problems. Throughout the semester, you and your group will engage in system development, research implementation, instrument validation, data gathering and analysis, technical





documentation, oral defense, project deployment, and institutional compliance. These activities simulate the professional responsibilities expected of future IT practitioners and researchers in the aviation industry.

This module serves as your roadmap for the entire semester. It provides a comprehensive orientation on the course requirements, grading system, consultation procedures, project timeline, ethical responsibilities, AI-use disclosure policies, and the expected deliverables that will guide your progress from project implementation to final completion. By understanding these expectations at the beginning of the course, you will be better prepared to organize your team, manage your time effectively, and accomplish every milestone according to institutional standards.

As you begin this module, remember that a successful capstone project is not only measured by a functional system but also by careful planning, effective teamwork, ethical research conduct, and professional commitment. Take this opportunity to establish a strong foundation that will support your success throughout the remainder of the course.

Lesson 1. Course Orientation

The instructor will introduce the overall objectives, scope, and expectations of Aviation Capstone Project 2. Students will become familiar with the course syllabus, grading policies, semester schedule, and institutional requirements. Orientation also includes an overview of the major milestones that every capstone group must accomplish throughout the semester.

During this lesson, learners are expected to understand the importance of project planning, teamwork, communication, and active participation in every consultation, research activity, and defense schedule.

Lesson 2. Capstone Project Roadmap

A capstone project is completed through a series of interconnected phases. Each phase builds upon the outputs produced in the previous stage. Understanding this workflow enables students to anticipate future tasks and manage their project efficiently.

The general roadmap of Aviation Capstone Project 2 includes:

- Course Orientation
- Proposal-Defense Output Review
- Development of Evaluation Instruments
- Validation and Reliability Testing
- Chapter IV Preparation
- Preliminary Examination





- Chapter V Writing
- Research-Day Consultation
- Mock Final Defense
- Final Defense
- Revision and Compliance
- Deployment
- Research Clearance

Each milestone represents an important requirement that contributes to the successful completion of the capstone project.

Lesson 3. Group Responsibilities

A capstone project is a collaborative undertaking that requires every member to contribute actively to the success of the group. During the first week, each group should establish a clear distribution of responsibilities based on the strengths, expertise, and availability of its members.

Typical responsibilities may include:

- Project Leader
- System Developer/Programmer
- Documentation Leader
- Database Administrator
- UI/UX Designer
- Quality Assurance and Testing Coordinator
- Research Coordinator
- Presentation Coordinator

Although individual roles may differ, every member shares collective responsibility for ensuring the quality and timely completion of the project.

Lesson 4. Consultation Protocol

Regular consultation with the adviser plays a crucial role in the successful completion of the capstone project. Consultation sessions provide opportunities to receive technical guidance, verify research progress, resolve project issues, and ensure compliance with institutional standards.

Students are expected to:

- Schedule consultations regularly.





- Come prepared with updated outputs.
- Record adviser recommendations.
- Revise outputs based on verified feedback.
- Maintain a consultation log throughout the semester.

Lesson 5. Ethics, Academic Integrity, and AI-use Disclosure

Research integrity is one of the fundamental principles of every capstone project. Students must ensure that all submitted outputs are original, properly cited, and free from plagiarism, fabrication, and falsification of data.

Artificial Intelligence (AI) tools may assist learners in improving grammar, generating programming suggestions, organizing ideas, or enhancing productivity. However, AI tools must never replace critical thinking, independent analysis, or original research. Any significant use of AI during the preparation of the manuscript or system development should be properly disclosed in accordance with institutional policies.

Students are likewise expected to uphold data privacy, respect intellectual property rights, and observe professional ethics throughout the entire research and system development process.

Lesson 6. Final Deliverables

At the end of the semester, every capstone group is expected to submit the required academic and technical outputs, including:

- Revised Research Manuscript
- Functional Information System
- Source Codes
- System Documentation
- Evaluation Results
- Validation Documents
- Consultation Log
- Revision Matrix
- Deployment Evidence
- Electronic Project Files
- Hardbound Research Manuscript
- Research Clearance Requirements

These deliverables serve as evidence that the capstone project has successfully satisfied both academic and institutional requirements.





References:

Books

Garcia, L. C., et al. (2016). **Outcomes-Based Practical Guide to Thesis and Capstone Project Writing in Computing**. Mindshapers Co., Inc.

Howell, D. C. (2008). **Fundamental Statistics for the Behavioral Sciences** (6th ed.).

Institutional References

Philippine State College of Aeronautics. (2025). *IT411: Aviation Capstone Project 2 course syllabus* (Version 2, Revision 1, 1st Semester, S.Y. 2025–2026). Institute of Computer Studies.

Philippine State College of Aeronautics. (2013). *PhilSCA student research handbook: AY 2013–2016*.





GUIDED LEARNING ACTIVITIES

Activity 1. Brainstorming: My Capstone Journey

Working with your group, discuss the possible challenges that you expect to encounter while completing Aviation Capstone Project 2. Identify at least five possible challenges and suggest practical strategies that your group can implement to overcome each challenge.

Rubric

Identification of Challenges – 30%

Practicality of Proposed Solutions – 30%

Participation and Collaboration – 20%

Clarity of Presentation – 20%

Activity 2. Think–Pair–Share: Understanding Team Responsibilities

Pair with another classmate and discuss the different responsibilities of each capstone member. After your discussion, share your ideas with the class and explain why teamwork is essential in completing a successful capstone project.

Rubric

Understanding of Assigned Topic – 40%

Quality of Discussion – 30%

Participation – 20%

Communication Skills – 10%

Activity 3. Concept Mapping: Capstone Project Roadmap

Construct a concept map that illustrates the complete workflow of Aviation Capstone Project 2, beginning with course orientation and ending with research clearance. Show the relationship among the major milestones discussed in this module.

Rubric

Accuracy of the Roadmap – 40%

Organization of Concepts – 30%

Visual Presentation – 20%

Creativity – 10%





FORMATIVE ASSESSMENT

Part A. Multiple Choice

Directions: Read each question carefully and choose the best answer.

1. What is the primary objective of Aviation Capstone Project 2?
2. Why is regular consultation with the adviser important?
3. Which document records the comments and recommendations provided by the adviser?
4. What is the purpose of AI-use disclosure?
5. Which of the following is NOT considered one of the major capstone deliverables?

Part B. Reflection

Write a short reflection (200–300 words) answering the following question:

How can proper planning, teamwork, and ethical research practices contribute to the successful completion of your capstone project?





PRELIMINARY PERIOD

Module 2

Proposal Defense Output and System / Documentation Presentation

Week: 2 | Time Allotment: 3 Hours

Learning Outcomes

Course Learning Outcomes (CLO)

This module contributes to the attainment of the following Course Learning Outcomes:

CLO 1. Evaluate approved capstone proposals and proposal-defense outputs using adviser/panel findings, routing documents, and institutional capstone requirements to determine readiness for system completion and final documentation with at least **80% compliance** based on the capstone readiness rubric.

CLO 8. Demonstrate professionalism, teamwork, ethical research practice, proper AI-tool disclosure, data privacy awareness, and respect for intellectual property throughout system development, defense, deployment, and clearance processes.

Module Learning Outcome (MLO)

MLO 2. Evaluate proposal-defense outputs, approved system direction, manuscript status, and pending improvements from Capstone Project 1.

Topic Learning Outcomes (TLO)

At the end of this module, learners should be able to:

TLO 1. Present the approved proposal-defense manuscript, system prototype, and documents previously submitted.

TLO 2. Analyze adviser and panel comments and prepare a revision matrix identifying system, documentation, and compliance gaps.

Learning Objectives

Upon successful completion of this module, the learners will be able to:

1. Present the approved proposal-defense outputs in an organized and professional manner.
2. Explain the current status of the capstone manuscript and system prototype.
3. Analyze adviser and panel recommendations from Capstone Project 1.
4. Differentiate completed outputs from those requiring revision.
5. Prepare a revision matrix that will serve as the basis for succeeding system development and manuscript improvement.
6. Demonstrate teamwork and professionalism during technical presentations and peer discussions.

Time Commitment

Learning Task	Estimated Time
Introduction	20 minutes
Learning Content	90 minutes
Guided Learning Activities	40 minutes
Formative Assessment	30 minutes
Total	3 Hours

Required Resources

Learners should prepare the following before attending the class:

- Approved proposal-defense manuscript
- System prototype
- Adviser and panel comments
- Routing slip
- Revision matrix template
- Google Slides or Canva presentation
- Laptop/Desktop computer
- Learning Management System (LMS)



- Internet connection

These resources are based on the Week 2 learning plan in the official course syllabus.

Module Overview

The successful completion of a proposal defense marks an important milestone in every capstone project; however, it is only the beginning of the implementation phase. The comments, recommendations, and observations provided by the adviser and panel members serve as valuable inputs for improving both the research manuscript and the proposed information system.

This module enables learners to revisit their approved proposal-defense outputs, evaluate the current status of their system prototype, identify areas requiring improvement, and organize the necessary revisions through a structured revision matrix. By examining these outputs critically, students establish a clear development direction that will guide the succeeding stages of system implementation and manuscript preparation.

Introduction

After successfully defending your capstone proposal, you are now entering the implementation phase of your research project. Although your proposal has been approved, this does not necessarily mean that every component is already complete or error-free. Most proposal defenses result in recommendations that strengthen the study, improve the system design, clarify the methodology, or enhance the quality of the manuscript.

Rather than viewing these recommendations as criticisms, they should be considered opportunities for improvement. Professional software developers and researchers continuously refine their work based on expert evaluation and stakeholder feedback. Likewise, capstone researchers are expected to document every recommendation, evaluate its impact on the project, and implement appropriate revisions before proceeding to system development.

This module emphasizes the importance of reviewing proposal-defense outputs objectively and systematically. Through presentations, peer review, and revision planning, learners will develop a clearer understanding of the work that has already been accomplished and the tasks that still need to be completed before moving to the next phase of the capstone project.





Lesson 1. Reviewing the Approved Proposal-Defense Outputs

Learning Focus

Before proceeding with the development of the proposed information system, every capstone group must first review the outputs generated during **Capstone Project 1**. These outputs serve as the official foundation of the implementation phase and provide the necessary direction for completing the remaining requirements of the research and system development process.

The proposal defense is not the conclusion of the capstone project; rather, it marks the transition from planning to implementation. During the proposal defense, students present their research problem, proposed methodology, system design, and implementation plan before a panel of evaluators. After evaluating the proposal, the panel may approve the study, recommend revisions, or require additional improvements before the project proceeds to development.

For this reason, reviewing the approved proposal-defense outputs is an essential first step in Aviation Capstone Project 2. This review enables students to understand the current status of their project, identify completed tasks, determine pending revisions, and establish a clear development roadmap for the remainder of the semester.

1.1 What are Proposal-Defense Outputs?

Proposal-defense outputs refer to the collection of academic, technical, and administrative documents produced and approved during **Capstone Project 1**. These outputs represent the official record of the group's proposed research and serve as the basis for all succeeding capstone activities.

Instead of starting from scratch, Capstone Project 2 continues the work that has already been approved. Therefore, every member of the group must become familiar with these documents before beginning system implementation.

- The primary proposal-defense outputs typically include:
- Approved research manuscript
- Proposal-defense presentation
- System prototype
- Adviser and panel recommendations
- Supporting documents

Each of these outputs plays a unique role in ensuring that the project progresses according to institutional standards and approved research objectives.



1.2 The Approved Research Manuscript

The **approved research manuscript** is one of the most important outputs carried over from Capstone Project 1. It serves as the official blueprint of the study and contains the approved research direction, system specifications, development plan, and documentation requirements that will guide the group throughout the implementation phase.

The manuscript is organized into the following major chapters:

- Chapter I – Introduction
- Chapter II – Review of Related Literature and Studies
- Chapter III – System Analysis and Design
- Chapter IV – Research Methodology, Results, and Discussion
- Chapter V – Summary, Conclusions, and Recommendations

At the beginning of Capstone Project 2, the manuscript may primarily contain the approved and revised sections from Chapters I to III. These sections establish the context of the project, define its purpose and objectives, present the relevant literature and studies, and describe the technical and methodological foundation of the proposed system.

Chapter I – Introduction includes the project context, project purpose and description, contextual framework, general and specific objectives, hypothesis when applicable, and scope and limitations.

Chapter II – Review of Related Literature and Studies presents the literature and previous studies related to the project using a thematic organization and IEEE citation format. It also includes a synthesis that explains how the reviewed sources support the current study.

Chapter III – System Analysis and Design contains the technical background, research and development methodology, software development life cycle model, system development framework, analysis of the existing system, feasibility study, cost-benefit analysis, system requirements specification, proposed system architecture, process and data models, security and data privacy design, and implementation environment and technology stack.

These chapters define what the project intends to accomplish, why the proposed system is needed, how it will be developed, and what technical and research requirements must be followed. They also establish the approved basis for evaluating whether the completed system remains aligned with the stated objectives and intended user requirements.

Since the manuscript has already undergone adviser and panel evaluation, students should not revise its content arbitrarily or change the approved project direction without proper consultation. Instead, they should carefully examine the panel recommendations, routing documents, and revision



notes to determine which sections require correction, clarification, expansion, or technical updating.

During Capstone Project 2, the approved manuscript will continue to be refined as the group completes system development, instrument validation, data gathering, system evaluation, and analysis of results. The approved Chapters I to III will serve as the foundation for preparing Chapter IV – Research Methodology, Results, and Discussion and Chapter V – Summary, Conclusions, and Recommendations.

Therefore, reviewing the approved manuscript is essential to ensure consistency among the research objectives, system features, development methodology, evaluation instruments, collected data, findings, conclusions, and recommendations.

1.3 The Proposal-Defense Presentation

The proposal-defense presentation is the slide deck used during the oral defense. Although often overlooked after the proposal defense, this presentation remains an important project document because it summarizes the essential components of the study.

- The presentation generally includes:
- Introduction
- Statement of the Problem
- Objectives of the Study
- Review of Related Literature
- Conceptual Framework
- Research Methodology
- Proposed System Architecture
- Project Timeline

Reviewing the presentation allows the group to verify that the content presented during the defense remains consistent with the approved manuscript. Any inconsistencies identified after the defense should be corrected to maintain accuracy throughout the project.

1.4 The System Prototype

The proposal defense usually includes an initial demonstration of the proposed information system. This may consist of wireframes, interface mock-





ups, database designs, process flowcharts, UML diagrams, or partially functional system modules.

The system prototype serves several important purposes.

First, it demonstrates the feasibility of the proposed solution.

Second, it allows advisers and panel members to evaluate whether the proposed system addresses the identified research problem.

Finally, it provides the development team with an initial design that will be expanded during the implementation phase.

During Module 2, students should examine whether the current prototype still aligns with the approved objectives of the study. Features that were recommended for revision or enhancement during the proposal defense should be identified and scheduled for development.

1.5 Adviser and Panel Recommendations

One of the most valuable outputs of the proposal defense is the collection of comments, observations, and recommendations provided by the adviser and panel members.

These recommendations may involve:

improving the research objectives;

refining the research methodology;

revising the system workflow;

adding or removing system features;

strengthening the literature review;

improving data collection procedures;

correcting formatting and citation issues; and

complying with institutional research requirements.

Rather than considering these recommendations as criticisms, students should recognize them as opportunities to improve the overall quality of the project. Addressing these comments demonstrates professionalism, academic maturity, and commitment to continuous improvement.

A best practice is to consolidate all recommendations into a **Revision Matrix**, where each comment is paired with the corresponding action taken and supporting evidence. This document enables advisers and panel members to verify that every recommendation has been addressed.



1.6 Supporting Documents

In addition to the manuscript and system prototype, capstone groups are expected to maintain complete supporting documents throughout the implementation process. These documents provide evidence that the project complies with institutional policies and research standards.

- Supporting documents may include:
- Proposal-defense approval sheet
- Routing forms
- Adviser consultation records
- Ethics clearance (if applicable)
- Research instruments
- Validation forms
- Gantt chart or project schedule
- Letters of cooperation or permission
- Initial system documentation

Maintaining organized documentation simplifies project monitoring and helps prevent delays during the final defense and research clearance.

1.7 Why Reviewing Proposal-Defense Outputs is Important

Reviewing proposal-defense outputs before beginning system development offers several important advantages.

First, it enables the group to identify completed and incomplete tasks. Instead of repeating work, students can focus on areas requiring improvement.

Second, it ensures consistency among the approved manuscript, presentation materials, and system prototype. Consistency minimizes errors and prevents conflicting information from appearing in different project documents.

Third, it helps the development team establish realistic priorities and timelines for the implementation phase.

Finally, it promotes accountability among group members by clarifying individual responsibilities and ensuring that all recommendations from the proposal defense are properly addressed.



Lesson 2. Understanding Adviser and Panel Recommendations

Learning Focus

One of the most valuable outcomes of a successful proposal defense is the collection of recommendations provided by the research adviser and panel members. These recommendations are intended to improve the overall quality of the capstone project by strengthening its research content, technical design, system functionality, and compliance with institutional standards.

Many students mistakenly believe that an approved proposal requires no further revisions. In reality, an approval often comes with comments, suggestions, and required modifications that must be addressed before the project proceeds to the implementation phase. These recommendations serve as a roadmap for refining the study and ensuring that the final output meets the expected academic and professional standards.

Understanding, documenting, and implementing these recommendations are essential responsibilities of every capstone group. Failure to address them may result in inconsistencies in the manuscript, technical deficiencies in the system, delays during consultations, or difficulties during the final defense.

2.1 Importance of Adviser and Panel Recommendations

The adviser and panel members evaluate the proposal from different professional perspectives. Their recommendations are based on their expertise in research, software engineering, information technology, system development, and academic writing.

Each recommendation aims to:

- improve the quality and clarity of the research;
- strengthen the technical design of the proposed system;
- ensure that the selected methodology is appropriate;
- verify that the proposed solution addresses the identified problem;
- ensure compliance with institutional, ethical, and academic standards; and
- increase the overall credibility and feasibility of the study.

Students should therefore regard every recommendation as an opportunity for improvement rather than as criticism. A well-documented revision process reflects professionalism, academic integrity, and a commitment to producing quality research.



2.2 Why Every Recommendation Should Be Documented

During the proposal defense, multiple comments may be provided by different evaluators. Some recommendations may require minor revisions, while others may involve significant modifications to the research manuscript or system design.

For this reason, every recommendation should be documented immediately after the defense. Proper documentation allows the research group to:

- avoid overlooking important corrections;
- organize revisions systematically;
- monitor the progress of completed modifications;
- provide evidence that adviser and panel recommendations have been addressed; and
- facilitate easier verification during subsequent consultations and the final defense.

A practical way of documenting recommendations is through a **Revision Matrix**. This document records each recommendation together with the corresponding action taken, the section of the manuscript or system affected, and the status of implementation.

Maintaining an updated revision matrix demonstrates that the group values constructive feedback and follows a systematic approach to project improvement.

2.3 Common Types of Adviser and Panel Recommendations

The recommendations received during the proposal defense may vary depending on the nature of the project. However, most comments generally fall into several common categories.

A. Project Context and Purpose

Panel members may recommend improving the discussion of the **Project Context** to provide a clearer explanation of the existing problem or operational situation that justifies the development of the proposed system.

Similarly, recommendations may focus on refining the **Project Purpose and Description** to ensure that the proposed solution clearly addresses the identified needs of the target users or organization.

Typical recommendations include:

- providing additional background information;



- strengthening the project justification;
- clarifying the intended beneficiaries; and
- ensuring alignment between the identified problem and the proposed solution.

B. Research Objectives

The research objectives define what the project intends to accomplish. Advisers and panel members often evaluate whether the general and specific objectives are clear, measurable, realistic, and aligned with the proposed system.

- Common recommendations include:
- revising vague or overly broad objectives;
- improving the sequence of objectives;
- ensuring consistency between objectives and system features; and
- removing objectives that are not achievable within the project timeline.

C. Research and Development Methodology

Another area frequently evaluated is the research and development methodology.

- Recommendations may include:
- justifying the selected Software Development Life Cycle (SDLC);
- improving the discussion of development phases;
- explaining why the chosen methodology is appropriate for the project;
- refining data gathering procedures; and
- strengthening the research design.

A clearly explained methodology increases the credibility and reliability of the study.

D. System Analysis and Design

Panel members also review the technical aspects of the proposed system. Recommendations may involve improvements to:

- system architecture;
- process workflows;
- context diagram;
- data flow diagrams;
- entity relationship diagram;
- database design;
- user interface design;
- system modules; and
- system security mechanisms.

These recommendations ensure that the system is technically sound, scalable, and capable of meeting user requirements.

E. System Features and Functional Requirements

The proposed system should include features that directly support the objectives of the study.

- Recommendations may involve:
- adding missing features;
- removing unnecessary functions;
- improving workflow efficiency;
- simplifying user interaction; and
- enhancing system usability.

Students should evaluate whether each recommended feature contributes to solving the identified problem.

F. Documentation and Academic Writing

Proposal-defense recommendations frequently address the quality of the research manuscript.

These may include:

- improving grammar and sentence construction;
- reorganizing sections;
- strengthening literature review;
- improving synthesis;

- correcting figure and table captions;
- updating references; and
- ensuring consistency throughout the manuscript.

Proper academic writing enhances the readability and professionalism of the research.

G. Formatting and Citation Requirements

Another common category involves formatting corrections.

Examples include:

- correcting headings and subheadings;
- applying consistent margins and spacing;
- revising page numbering;
- improving tables and figures;
- following the prescribed manuscript format; and
- ensuring proper citation style and reference formatting.

Although formatting revisions may appear minor, they contribute significantly to the overall quality and professionalism of the manuscript.

H. Compliance and Supporting Documents

In addition to the manuscript and system design, advisers and panel members may identify missing institutional requirements.

- These may include:
- consultation records;
- routing forms;
- ethics-related documents;
- validation forms;
- letters of cooperation;
- AI-use disclosure;
- plagiarism reports; and
- other supporting documents required by the institution.

Ensuring complete compliance helps prevent delays during project evaluation and final clearance.

2.4 Responding Professionally to Recommendations

Receiving numerous recommendations should not discourage a research group. Instead, students should view the proposal defense as a collaborative process in which experts help improve the quality of the project.

A professional research team should:

- listen attentively during the proposal defense;
- document every recommendation accurately;
- seek clarification whenever necessary;
- discuss recommendations with the adviser;
- implement revisions systematically;
- update the revision matrix regularly; and
- present revised outputs during succeeding consultations.

Professionalism is demonstrated not by avoiding mistakes but by responding positively to constructive feedback and continuously improving the quality of the project.

Best Practices in Managing Recommendations

To ensure that no recommendation is overlooked, every capstone group should adopt the following practices:

1. Assign one member to document all adviser and panel comments during the proposal defense.
2. Consolidate all recommendations into a single revision matrix immediately after the defense.
3. Prioritize revisions according to urgency and their impact on the project.
4. Verify completed revisions with the adviser before proceeding to the next development phase.
5. Keep both printed and digital copies of all revised documents for future reference and compliance.

Following these practices promotes efficient project management and minimizes delays during consultations and the final defense.

Lesson 3. Evaluating the Current System Prototype

Learning Focus

After reviewing the approved proposal-defense outputs and understanding the recommendations provided by the adviser and panel members, the next step in the implementation phase is to evaluate the current system prototype. This process enables the development team to determine whether the existing prototype remains aligned with the approved research objectives, user requirements, and technical specifications presented during the proposal defense.

A system prototype represents the initial version of the proposed information system. Although it may not yet be fully functional, it provides a visual and technical representation of how the final system is expected to operate. Evaluating the prototype helps identify strengths, weaknesses, missing features, usability concerns, and technical improvements before the system proceeds to full-scale development.

Rather than simply demonstrating what has already been developed, prototype evaluation encourages critical analysis and continuous improvement. It allows the research group, adviser, classmates, and other stakeholders to verify whether the proposed system effectively addresses the identified problem and satisfies the needs of its intended users.

3.1 What is a System Prototype?

A **system prototype** is an early version of an information system that illustrates its intended functionality, user interface, workflow, and major system components. It serves as a working model that enables developers, advisers, and users to visualize how the completed system will operate.

Depending on the stage of development, a prototype may include:

- User interface mock-ups
- Wireframes
- Navigation menus
- Sample data
- Initial database structure
- Partially functional modules
- Demonstration of selected system features

The prototype is not necessarily the finished product. Instead, it is a developmental tool used to collect feedback, verify requirements, and identify necessary improvements before the final implementation.

3.2 Purpose of Prototype Evaluation

Prototype evaluation is conducted to determine whether the current version of the system remains consistent with the approved proposal and continues to address the objectives of the study.

Specifically, prototype evaluation aims to:

determine whether the developed features satisfy the approved research objectives;

verify that user requirements have been properly translated into system functions;

- identify missing or incomplete modules;
- assess the usability of the system interface;
- evaluate the logical flow of system processes;
- detect technical errors before full implementation; and
- gather constructive feedback for continuous improvement.

Evaluating the prototype at an early stage reduces development risks, minimizes costly revisions later, and improves the overall quality of the completed information system.

3.3 Evaluating Alignment with the Research Objectives

One of the first considerations during prototype evaluation is determining whether the system directly supports the objectives identified in the approved manuscript.

Each specific objective should correspond to one or more system functions. If a particular objective cannot be demonstrated through the prototype, the development team should determine whether the feature has not yet been developed or whether revisions to the system design are necessary.

For example, if one of the objectives is to **automate student internship monitoring**, the prototype should demonstrate features such as:

- student registration;
- internship company management;
- attendance monitoring;
- progress tracking; and
- report generation.

When every objective has a corresponding system function, the project demonstrates strong alignment between the research component and the technical solution.

3.4 Evaluating System Functionality

Functionality refers to the ability of the system to perform the tasks for which it was designed.

During evaluation, the development team should verify whether the implemented modules perform their intended functions correctly.

Examples of questions that may be considered include:

- Does the login module authenticate users properly?
- Can users successfully add, edit, delete, and retrieve records?
- Are reports generated accurately?
- Do all menu options function correctly?
- Are error messages displayed appropriately?

Functional evaluation focuses on whether the system performs correctly and produces the expected outputs under normal operating conditions.

3.5 Evaluating the User Interface Design

The user interface (UI) is one of the first aspects noticed by users. An effective interface should be visually appealing, organized, consistent, and easy to navigate.

During evaluation, the following elements should be examined:

- layout consistency;
- readability of text;
- appropriate use of colors;
- icon and button placement;
- navigation efficiency;
- responsiveness of pages; and
- accessibility for intended users.

A well-designed interface contributes to a positive user experience and improves overall system usability.

3.6 Evaluating the System Workflow

Workflow refers to the sequence of activities performed by users while interacting with the system.

A logical workflow enables users to complete tasks efficiently without unnecessary complexity.

For example, in an aviation maintenance management system, the expected workflow may include:

- User authentication
- Dashboard access
- Maintenance request creation
- Supervisor approval
- Technician assignment
- Maintenance completion
- Report generation

During evaluation, developers should determine whether every process follows a logical sequence and whether users can accomplish tasks efficiently.

3.7 Evaluating System Completeness

Another important aspect of prototype evaluation is determining the completeness of the system.

- Students should compare the implemented prototype with:
 - approved objectives;
 - proposed system features;
 - functional requirements;
 - approved modules; and
 - project timeline.

This comparison helps identify which components have already been completed and which remain under development.

Incomplete modules should be documented and incorporated into the group's implementation schedule.

3.8 Receiving Constructive Feedback

Prototype evaluation is most effective when multiple perspectives are considered. During classroom presentations, advisers, instructors, classmates, and other evaluators may provide constructive observations regarding the proposed system.

- Constructive feedback may involve:
- improving system functionality;
- simplifying navigation;
- enhancing interface design;
- strengthening security;
- correcting workflow issues;
- adding missing reports;
- improving data validation; and
- enhancing user experience.

Students should document every suggestion and evaluate whether it contributes to the overall objectives of the project.

Receiving constructive criticism is an essential part of professional software development. Successful developers continuously improve their systems based on user feedback and expert recommendations.

3.9 Best Practices During Prototype Presentation

When presenting the current system prototype, each capstone group should observe the following practices:

Prepare the system before the presentation to avoid technical interruptions.

Demonstrate only the implemented features and clearly identify modules that are still under development.

- Explain how each feature supports the research objectives.
- Present realistic sample data whenever possible.
- Record all observations provided by the instructor and classmates.
- Respond professionally to questions and suggestions.
- Update the revision matrix after the presentation.



Professional presentations demonstrate not only technical competence but also preparedness, teamwork, and confidence in the proposed solution.

Lesson 4. Preparing the Revision Matrix

Learning Focus

One of the most important responsibilities of every capstone research group after the proposal defense is to ensure that all recommendations provided by the research adviser and panel members are properly addressed. These recommendations may involve revisions to the research manuscript, system design, software features, documentation, research methodology, or institutional requirements. To organize and monitor these revisions effectively, capstone groups prepare a **Revision Matrix**.

A revision matrix is more than a simple checklist of corrections. It is a project management document that systematically records every recommendation received during consultations and proposal defense presentations, the actions taken by the research group, the evidence supporting the revisions, and the verification provided by the adviser. It serves as an official record of project improvements and demonstrates that the group has carefully considered and implemented the feedback received from evaluators.

Maintaining an updated revision matrix promotes accountability, facilitates communication between the research group and the adviser, and provides documented evidence that revisions have been completed throughout the capstone process.

4.1 What is a Revision Matrix?

A **Revision Matrix** is a structured document used to record, monitor, and verify all revisions made to a capstone project based on the comments and recommendations of the research adviser, panel members, and other evaluators.

It serves as a centralized record that enables the research group to track project improvements from the proposal stage until the completion of the final manuscript and information system.

Rather than relying on memory or handwritten notes, students can systematically document every recommendation and monitor its implementation throughout the semester.

The revision matrix also provides transparency during consultations because advisers can immediately determine which recommendations have already been completed and which still require further work.



4.2 Purpose of the Revision Matrix

The primary purpose of the revision matrix is to ensure that every recommendation received during the proposal defense and succeeding consultations is properly addressed.

Specifically, the revision matrix helps the research group to:

- organize adviser and panel recommendations systematically;
- prioritize revisions according to their importance;
- monitor the progress of completed and pending revisions;
- maintain consistency between the research manuscript and the developed system;
- provide documented evidence of project improvements; and
- facilitate easier verification during adviser consultations and final defense.

Because multiple consultations take place throughout Capstone Project 2, the revision matrix becomes an important project management tool that records the development history of the research.

4.3 Components of a Revision Matrix

Although revision matrices may vary among institutions, a standard capstone revision matrix generally contains the following information:

Column	Description
Item No.	Sequential number assigned to each recommendation.
Source of Recommendation	Indicates whether the comment came from the adviser, panel member, statistician, grammar reviewer, or another evaluator.
Recommendation	The exact recommendation or observation provided during the proposal defense or consultation.
Affected Section	Identifies the chapter, section, or system module requiring revision.
Action Taken	Describes how the recommendation was addressed by the research group.

Supporting Evidence	Indicates the page number, screenshot, figure, table, revised paragraph, or system module that demonstrates the completed revision.
Status	Indicates whether the recommendation is Completed, Ongoing, or Pending.
Verified By	Adviser or evaluator who confirms that the revision has been satisfactorily addressed.

Including these components helps maintain an organized and transparent revision process throughout the project.

4.4 How to Prepare a Revision Matrix

Preparing a revision matrix begins immediately after the proposal defense or adviser consultation.

The following procedure is recommended:

Step 1. Gather all recommendations.

Collect every comment provided by the adviser, panel members, and other evaluators. Review consultation notes, evaluation sheets, proposal-defense forms, and personal notes taken during the presentation.

Step 2. Record each recommendation.

Write every recommendation exactly as provided by the evaluator. Avoid changing its meaning or combining multiple recommendations into one statement.

Step 3. Identify the affected section.

Determine which chapter, subsection, figure, table, diagram, or system module is affected by the recommendation.

Examples include:

- Chapter I – Project Context
- Chapter II – Review of Related Literature
- Chapter III – Functional Requirements
- Login Module
- Dashboard Interface



- Entity Relationship Diagram

Step 4. Describe the action taken.

After implementing the recommendation, describe the specific revision that was made.

Examples include:

- Revised the Project Context by incorporating current industry statistics.
- Updated the Functional Requirements to include administrator approval.
- Modified the login module to support multi-factor authentication.
- Redesigned the dashboard layout for improved usability.
- The action taken should clearly explain what was changed.

Step 5. Attach supporting evidence.

Every completed revision should be supported by evidence.

Examples include:

- Revised page number
- Updated screenshot
- Figure number
- Source code reference
- Database diagram
- Consultation output
- Supporting evidence allows advisers to verify revisions more efficiently.

Step 6. Update the status.

The status column should accurately reflect the progress of every recommendation.

Common status indicators include:

- Completed – Revision has been fully implemented.
- Ongoing – Revision is currently being implemented.
- Pending – Revision has not yet been started.



- The revision matrix should be updated regularly after every consultation.

4.5 Sample Revision Matrix

The following table illustrates a simple example of a capstone revision matrix.

Item	Source	Recommendation	Affected Section	Action Taken	Evidence	Status
1	Panel Member 1 Name	Clarify the Project Context by adding current aviation industry data.	Chapter I – Project Context	Added recent aviation statistics and supporting references.	Page 4	Completed
2	Adviser Name	Include password encryption in the system design.	Chapter III – Security Design	Implemented password hashing and updated the security design discussion.	Figure 3.15	Completed
3	Panel Member 2 Name	Improve dashboard navigation.	Dashboard Module	Rearranged menu items and simplified navigation flow.	Screenshot 5	Ongoing

This matrix provides a clear record of every recommendation, the actions taken by the group, and the current implementation status.

4.6 Best Practices in Maintaining a Revision Matrix

To maximize the usefulness of the revision matrix, capstone groups should observe the following practices:

- Update the matrix immediately after every adviser consultation or proposal defense.
- Record recommendations accurately without changing their original intent.
- Provide concise but complete descriptions of the revisions made.
- Attach sufficient evidence to support every completed revision.
- Review the matrix regularly before each consultation.
- Retain both printed and digital copies of the latest revision matrix.

Maintaining an updated revision matrix demonstrates professionalism, organization, and commitment to continuous project improvement.



4.7 Importance of the Revision Matrix Throughout the Capstone Process

The revision matrix is not only useful immediately after the proposal defense. It remains an essential document throughout Capstone Project 2 because it serves as a historical record of the project's development.

As the research progresses, the matrix helps the research group:

- monitor improvements made to the manuscript and information system;
- ensure consistency between adviser recommendations and implemented revisions;
- prepare for consultation meetings;
- facilitate the verification of completed revisions during the final defense;
- and
- provide documented evidence of continuous quality improvement.

For these reasons, many institutions require the revision matrix to be presented during consultations and, in some cases, submitted together with the revised manuscript as part of the project documentation.

Lesson 5. Identifying Documentation and Compliance Gaps

Learning Focus

The successful completion of a capstone project requires more than developing a functional information system or writing a well-structured research manuscript. Every research group must also ensure that all required documents, institutional requirements, and supporting records are complete, organized, and available throughout the capstone process. Failure to comply with documentation and institutional requirements may delay consultations, defense schedules, manuscript approval, deployment, or even the final research clearance.

After the proposal defense, research groups should conduct a thorough review of all project documents to identify missing, incomplete, or outdated requirements. This process is known as **documentation and compliance gap identification**. By identifying these gaps early, the research group can prepare an action plan to complete all outstanding requirements before moving to the succeeding phases of the project.

Effective documentation management reflects professionalism, accountability, and organizational skills—qualities expected of future information technology professionals and researchers.



5.1 Understanding Documentation and Compliance

In every capstone project, **documentation** refers to the collection of academic, technical, administrative, and supporting records that provide evidence of the project's development and implementation.

Compliance, on the other hand, refers to the fulfillment of institutional, academic, ethical, and technical requirements prescribed by the college or university before a project can proceed to the next stage.

While documentation focuses on maintaining accurate project records, compliance ensures that all institutional policies, procedures, and standards have been properly observed.

Together, documentation and compliance provide assurance that the capstone project has been conducted systematically, ethically, and according to institutional requirements.

5.2 Why Documentation is Important

Proper documentation serves several important purposes throughout the capstone project.

It enables the research group to:

- maintain organized records of project activities;
- monitor project progress and completed milestones;
- provide evidence during adviser consultations;
- support system validation and evaluation;
- facilitate manuscript revision and verification;
- prepare for proposal, mock, and final defense; and
- complete institutional clearance requirements efficiently.

Without proper documentation, valuable project information may be lost, revisions may become difficult to trace, and important institutional requirements may be overlooked.

5.3 Common Documentation Required in Capstone Projects

Although institutional requirements may vary, most capstone projects require the preparation and maintenance of the following documents:

Research Documents

- Approved research manuscript
- Revised manuscript versions
- Proposal-defense presentation
- Revision matrix
- Research instruments
- Validation forms
- Statistical analysis outputs
- References and bibliography

Technical Documents

- Software Requirements Specification (SRS)
- System architecture
- Context Diagram
- Data Flow Diagram (DFD)
- Entity Relationship Diagram (ERD)
- Database design
- User Interface designs
- Source code documentation
- User Manual
- Technical Documentation

Administrative Documents

- Consultation log
- Adviser consultation forms
- Proposal-defense evaluation sheets
- Routing forms
- Approval sheets
- Progress monitoring forms
- Attendance records (if required)

Supporting Documents

- Letters of cooperation
- Permission letters
- Ethics-related documents (if applicable)
- AI-use disclosure
- Plagiarism report (if required)
- System screenshots
- Deployment evidence
- Acceptance forms
- Presentation materials

Maintaining these documents in an organized manner reduces the possibility of missing important requirements during project evaluation.

5.4 Identifying Documentation Gaps

A **documentation gap** exists when a required document is incomplete, outdated, inaccurate, or entirely missing.

Some common documentation gaps include:

- incomplete manuscript chapters;
- missing signatures or approval pages;
- outdated system diagrams;
- incomplete references;
- absent consultation records;
- missing validation forms;
- undocumented system revisions; and
- incomplete technical documentation.

These gaps should be identified as early as possible to prevent delays during consultations and defense schedules.

5.5 Identifying Compliance Gaps

A **compliance gap** occurs when the project fails to satisfy one or more institutional requirements.

Examples include:

- incomplete routing requirements;
- missing adviser approval;
- failure to submit required reports on time;
- incomplete revision matrix;
- incomplete deployment evidence;
- missing AI-use disclosure;
- failure to comply with manuscript formatting requirements;
- incomplete supporting documents required for research clearance.

Unlike documentation gaps, compliance gaps involve institutional procedures that must be completed before the project can advance to the next stage.

5.6 Conducting a Documentation and Compliance Audit

One effective strategy for identifying project gaps is to conduct a **documentation and compliance audit**.

This process involves reviewing every project document using a standardized checklist.

- The research group should verify:
- Is every required document available?
- Is the latest version being used?
- Has every adviser recommendation been incorporated?
- Are all signatures complete?
- Are supporting documents attached?
- Are technical documents updated?
- Are all institutional forms completed?

Conducting regular audits allows the group to detect missing requirements before scheduled consultations and defense activities.

5.7 Organizing Project Documents

Proper organization is essential for efficient project management.

Capstone groups are encouraged to maintain both **printed** and **digital** copies of all project documents.

A recommended digital folder structure is shown below.

CAPSTONE PROJECT

- Chapter 1
- Chapter 2
- Chapter 3
- Chapter 4
- Chapter 5
- System Source Code
- Database
- Research Instruments
- Consultation Logs
- Revision Matrix
- Technical Documents
- Presentation Slides
- Deployment Documents
- Supporting Documents
- Final Submission

Using a standardized folder structure allows every member of the research group to locate documents quickly and minimizes the risk of losing important project files.

5.8 Best Practices in Documentation Management

To maintain complete and organized project records, research groups should observe the following best practices:

- Update project documents immediately after every consultation.
- Save digital files using consistent filenames and version numbers.
- Maintain cloud backups using institutional storage platforms.
- Archive previous manuscript versions before making revisions.
- Assign one member to monitor documentation completeness.
- Review compliance requirements regularly with the adviser.
- Maintain confidentiality of research data and sensitive project information.



These practices contribute to better project management and reduce the likelihood of missing important requirements.

5.9 Preparing for the Next Phase of the Capstone Project

Before proceeding to the succeeding modules, every capstone group should verify that all documentation and compliance requirements are complete.

The project should have:

- an updated research manuscript;
- an organized revision matrix;
- complete consultation records;
- updated technical documentation;
- complete supporting documents;
- verified institutional forms; and
- properly organized digital and printed project files.

Completing these preparations establishes a strong foundation for the remaining phases of system development, research implementation, evaluation, deployment, and final defense.

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Guided Learning Activities

Activity 1. Simulation and Demonstration

Each capstone group presents the following:

- Approved proposal-defense manuscript
- Current system prototype
- Supporting project documents

The presentation should highlight the current status of the project, major accomplishments, and pending improvements.

Activity 2. Peer Teaching and Review

Groups exchange presentations and evaluate one another using a structured peer feedback form. Constructive comments should focus on system functionality, documentation quality, and presentation clarity.

Activity 3. Gallery Walk

Each group displays its proposal-defense outputs, adviser comments, routing documents, and draft revision matrix. Students rotate among the displays, examine the presented outputs, and provide written suggestions for further improvement.

Formative Assessment

Part A. Reflection Questions

Answer the following briefly:

1. Why is it important to review the comments provided during the proposal defense?
2. How does a revision matrix help improve a capstone project?
3. What documents should be reviewed before beginning system implementation?
4. Why is peer feedback valuable in improving research quality?
5. How can proper documentation contribute to the successful completion of a capstone project?

Part B. Submission Requirements



By: ASSOC. PROF. MARY ANN F. ABALLIAR VISTA



1st Semester,
SY 2025-2026

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Each group shall submit:

1. Updated proposal-defense presentation
2. Initial revision matrix
3. Annotated adviser and panel comments
4. Updated system prototype status report

Rubrics

Rubric A. Proposal Defense Output and System/Document Presentation

Criteria	Weight
Organization and completeness of proposal-defense documents	25%
Quality and functionality of the system prototype	25%
Analysis of adviser and panel recommendations	20%
Accuracy and completeness of the revision matrix	20%
Professionalism, teamwork, and presentation delivery	10%
Total	100%

REFERENCES:

Institutional References

1. National Aviation Academy of the Philippines. (2026). **IT411: Aviation Capstone Project 2 Course Syllabus** (Version 2, Revision 2).
2. Philippine State College of Aeronautics. (n.d.). **Capstone Research Manual**.

Books and Writing References

3. American Psychological Association. (2020). **Publication manual of the American Psychological Association** (7th ed.).



PRELIMINARY PERIOD

Module 3

System Presentation

Week: 3 | Time Allotment: 3 Hours

Module Learning Outcome (MLO 3)

Demonstrate readiness of the system in the real world.

Target Learning Outcomes (TLO)

At the end of this module, the students should be able to:

- **TLO 04:** Present the completed system prototype.
- **TLO 10:** Collaborate with peers to showcase project progress.

Learning Objectives

At the end of this module, the students should be able to:

1. prepare a functional system prototype for presentation;
2. demonstrate the major features and functions of the developed system;
3. organize an effective presentation using appropriate visual aids;
4. receive and document comments from faculty members and peers; and
5. improve both the system and manuscript based on the feedback received.

Introduction

The System Presentation serves as an important milestone in Capstone Project 2. It allows student researchers to demonstrate the current progress of their developed system before faculty members and classmates. The presentation provides an opportunity to verify whether the prototype satisfies the approved project objectives and whether it is ready for system evaluation and data gathering.

Constructive feedback gathered during the presentation enables the research team to identify areas for improvement before proceeding to the next stages of the project. The comments and recommendations collected during this activity should also be





reflected in the corresponding revisions of both the manuscript and the system prototype.

Preparing the Prototype

Before the presentation, the research team should ensure that the system prototype is functional and ready for demonstration. The prototype should represent the approved design presented during Capstone Project 1 and should include the major features identified in the research objectives.

Prior to the presentation, verify the following:

- The system can be launched without errors.
- User accounts are available for demonstration.
- Sample data are prepared.
- Required modules are functional.
- Reports and outputs can be generated.
- Internet connection and software dependencies are available, if applicable.

Preparing the prototype before the presentation minimizes technical interruptions and allows the group to focus on demonstrating the capabilities of the developed system.

System Demonstration

A system demonstration is the process of presenting the functionality of the developed system to faculty members and classmates. During the demonstration, the presenters should explain how the system addresses the identified problem and satisfies the approved research objectives.

A typical demonstration may include:

- Introduction of the project
- Brief discussion of the system objectives
- User login
- Navigation of the major modules
- Sample transaction or process
- Report generation
- Summary of system features
- Questions and Answers

The demonstration should emphasize how the developed system operates in a realistic environment and how it provides a solution to the identified problem.

Preparing Presentation Slides



Presentation slides should summarize the important components of the project without reproducing the entire manuscript. Slides should be clear, organized, and visually appealing to help the audience understand the presentation.

The presentation may include the following sections:

- Title of the Study
- Background of the Study
- Research Problem
- Objectives of the Study
- Scope and Delimitation
- System Architecture or Framework
- System Features
- Prototype Demonstration
- Initial Accomplishments
- Next Project Milestones

Visual elements such as screenshots, diagrams, workflow illustrations, and system interfaces may be used to enhance the presentation.

Faculty Feedback

Faculty feedback is an essential part of the System Presentation. The comments provided by the instructor and classmates help identify strengths, weaknesses, and opportunities for improvement.

Feedback commonly focuses on:

- completeness of the prototype;
- consistency between the system and research objectives;
- usability and navigation;
- presentation organization;
- technical accuracy; and
- recommendations for system enhancement.



The research team should carefully document all comments received and determine the corresponding revisions that need to be incorporated into the manuscript and the prototype before the next module

Module Summary

Module 3 enables students to present the current version of their system prototype and receive constructive feedback from faculty members and peers. Through the preparation of the prototype, system demonstration, presentation slides, and faculty consultation, students evaluate the readiness of their project before proceeding to system evaluation and data gathering. The feedback gathered during this activity serves as an important reference for improving both the developed system and the research manuscript.

References:

International Organization for Standardization. (2023). **ISO/IEC 25010:2023: Systems and software engineering—Systems and software Quality Requirements and Evaluation (SQuaRE)—Product quality model.** <https://www.iso.org/standard/78176.html>

Davis, F. D. (1989). **Perceived usefulness, perceived ease of use, and user acceptance of information technology.** *MIS Quarterly*, 13(3), 319–340. <https://doi.org/10.2307/249008>

Davis, F. D., Bagozzi, R. P., & Warshaw, P. R. (1989). **User acceptance of computer technology: A comparison of two theoretical models.** *Management Science*, 35(8), 982–1003. <https://doi.org/10.1287/mnsc.35.8.982>

Online Resources

1. Commission on Higher Education (CHED): <https://ched.gov.ph>
2. IEEE Xplore Digital Library: <https://ieeexplore.ieee.org>
3. ACM Digital Library: <https://dl.acm.org>
4. Google Scholar: <https://scholar.google.com>



Learning Activity

Activity 3: Group System Presentation

Objectives

At the end of this activity, each Capstone Project group should be able to:

1. demonstrate the functionality of the developed system;
 2. explain how the system satisfies the research objectives;
 3. collaborate effectively during the presentation; and
 4. document the recommendations received from faculty members and peers.
-

Directions: Each Capstone Project group shall conduct a 15–20-minute System Presentation.

The presentation should include the following:

- Project Title
- Background of the Study
- Objectives of the Study
- Brief Description of the Developed System
- System Architecture or Workflow
- Demonstration of Major Modules
- Sample Transactions
- Generated Reports or Outputs
- Current Project Status
- Questions and Answers

Following the presentation, the group shall receive comments and recommendations from faculty members and classmates.



Documentation of Feedback

Record the comments received during the presentation.

Reviewer	Recommendation	Action to be Taken
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This documentation will serve as a reference for improving the system prototype and revising the manuscript.

Assessment

The following outputs shall be evaluated for Module 3.

Output	Assessment Tool
Group System Presentation	Presentation Rubric
Peer Review	Peer Review Form
Annotated Manuscript	Manuscript Evaluation Checklist

Required Submission

Before the end of Week 3, each group shall submit the following:

- Updated Presentation Slides
- Current System Prototype
- Peer Review Form

Annotated Manuscript reflecting comments and recommendations received during the presentation



PRELIMINARY PERIOD

MODULE 4

Survey Questionnaire Formulation

Week: 4 | Time Allotment: 3 Hours

Module Learning Outcome (MLO 4)

Develop appropriate survey questionnaires for evaluating the developed system and gathering research data.

Target Learning Outcomes (TLO)

At the end of this module, the students should be able to:

- formulate appropriate survey questionnaires for the identified respondents;
- construct evaluation instruments based on the objectives of the study;
- apply the ISO/IEC 25010 Software Quality Model in preparing system evaluation questionnaires;
- identify adopted survey instruments appropriate to the study; and
- validate the developed survey questionnaire prior to data gathering.

Learning Objectives

At the end of this module, the students should be able to:

1. explain the purpose of survey questionnaires in research;
 2. construct questionnaires aligned with the research objectives;
 3. prepare an ISO/IEC 25010-based evaluation questionnaire;
 4. identify appropriate adopted survey instruments from related studies; and
 5. prepare the questionnaire for expert validation.
-





Introduction

A survey questionnaire is one of the most commonly used instruments for collecting research data. In Capstone Project 2, questionnaires are developed to obtain responses from target users, IT experts, and other respondents who will evaluate the developed system.

A well-designed questionnaire ensures that the collected data are valid, reliable, and directly related to the objectives of the study. Before actual data gathering, the questionnaire should undergo validation to determine whether the items are clear, relevant, and appropriate for the intended respondents.

This module guides students in developing survey questionnaires using recognized evaluation standards and validated research instruments.

Survey Questionnaire

A survey questionnaire is a structured research instrument composed of questions designed to gather information from respondents. It allows researchers to collect quantitative or qualitative data that will answer the research objectives.

In Capstone Project 2, survey questionnaires are commonly used to:

- evaluate the developed system;
- determine user satisfaction;
- measure software quality;
- assess system usability; and
- obtain recommendations for system improvement.

A questionnaire should be:

- clear and understandable;
- objective;
- relevant to the research objectives;
- free from ambiguity; and
- appropriate for the intended respondents.

Questionnaire Construction

Questionnaire construction involves designing questions that accurately measure the variables of the study.

When constructing a questionnaire, researchers should consider the following:



The questionnaire should be reviewed carefully before validation to eliminate unclear or redundant questions.

Types of Questionnaires

ISO/IEC 25010

The **ISO/IEC 25010:2023** standard provides a framework for evaluating the quality of software products. It identifies nine (9) product-quality characteristics that may be used to assess whether a developed system meets the required quality standards. These characteristics serve as the basis for constructing evaluation questionnaires and measuring the overall quality of an information system.

The nine product-quality characteristics are described as follows:

1. Functional Suitability

Functional suitability refers to the degree to which the system provides the functions necessary to meet the specified requirements. It evaluates whether the software delivers accurate, complete, and appropriate results for the intended users and tasks.

2. Performance Efficiency

Performance efficiency measures how effectively the system utilizes computing resources while maintaining acceptable response times under normal operating conditions. It considers processing speed, resource utilization, and system capacity.

3. Compatibility

Compatibility refers to the ability of the system to coexist, communicate, and exchange information effectively with other software applications, hardware devices, or operating environments without affecting their performance.

4. Interaction Capability

Interaction capability evaluates the extent to which users can understand, learn, operate, and interact with the system efficiently and effectively. It emphasizes user interface design, ease of learning, accessibility, and overall user experience.

5. Reliability

Reliability measures the ability of the system to perform its intended functions consistently under specified conditions. It also evaluates the system's capability to recover from failures while maintaining the integrity of data and services.

6. Security

Security refers to the protection of information and system resources against unauthorized access, disclosure, modification, destruction, or misuse. It also ensures that users can perform only the functions authorized for their assigned roles.

7. Maintainability

Maintainability measures the ease with which the system can be analyzed, modified, tested, updated, corrected, and enhanced throughout its operational life cycle.

8. Flexibility

Flexibility refers to the ability of the system to adapt to changing requirements, operating environments, technologies, or organizational needs. It also considers the ease of installation, configuration, replacement, and scalability.

9. Safety

Safety evaluates the extent to which the system minimizes or prevents conditions that could cause harm to people, property, data, equipment, or the environment during its operation.

Adopted Survey Instruments

Rather than creating all questionnaire items from scratch, researchers may adopt or adapt existing survey instruments from published studies, journal articles, or internationally accepted standards.

When adopting an instrument, researchers should ensure that:

- the source is properly cited;
- the instrument is relevant to the objectives of the study;
- modifications are clearly identified; and
- permission is obtained when necessary.

Adopting validated instruments helps improve the reliability and credibility of the research.

Questionnaire Validation

Before administering the questionnaire to respondents, it should undergo validation.

Validation determines whether the questionnaire measures what it is intended to measure.

The questionnaire may be evaluated by:

- Subject Matter Experts
- IT Experts
- Research Experts
- Language Experts
- Adviser

Validators examine the questionnaire for:

- relevance;
- clarity;
- organization;
- completeness; and



- appropriateness of the items.

Based on the validators' recommendations, the questionnaire should be revised before data gathering.

Module Summary

Module 4 focuses on the development of survey questionnaires that will be used to evaluate the developed system and gather research data. Students learn the principles of questionnaire construction, apply the ISO/IEC 25010 Software Quality Model, identify appropriate adopted survey instruments, and prepare the questionnaire for expert validation. A properly designed and validated questionnaire contributes to the accuracy and credibility of the research findings.

References:

Books

Garcia, L. C., et al. (2016). **Outcomes-Based Practical Guide to Thesis and Capstone Project Writing in Computing**. Mindshapers Co., Inc.

Howell, D. C. (2008). **Fundamental Statistics for the Behavioral Sciences** (6th ed.).

Institutional References

Philippine State College of Aeronautics. (2025). *IT411: Aviation Capstone Project 2 course syllabus* (Version 2, Revision 1, 1st Semester, S.Y. 2025–2026). Institute of Computer Studies.

Philippine State College of Aeronautics. (2013). *PhilSCA student research handbook: AY 2013–2016*.

Standards:

International Organization for Standardization. (2023). **ISO/IEC 25010:2023: Systems and software engineering—Systems and software Quality Requirements and Evaluation (SQuaRE)—Product quality model**. <https://www.iso.org/standard/78176.html>





- Davis, F. D. (1989). **Perceived usefulness, perceived ease of use, and user acceptance of information technology.** *MIS Quarterly*, 13(3), 319–340. <https://doi.org/10.2307/249008>
- Davis, F. D., Bagozzi, R. P., & Warshaw, P. R. (1989). **User acceptance of computer technology: A comparison of two theoretical models.** *Management Science*, 35(8), 982–1003. <https://doi.org/10.1287/mnsc.35.8.982>

Online Resources

1. Commission on Higher Education (CHED): <https://ched.gov.ph>
2. IEEE Xplore Digital Library: <https://ieeexplore.ieee.org>
3. ACM Digital Library: <https://dl.acm.org>
4. Google Scholar: <https://scholar.google.com>





Learning Activity

Activity 4: Survey Questionnaire Development and Validation

Objectives

At the end of this activity, each Capstone Project group should be able to:

- develop a survey questionnaire aligned with the research objectives;
- prepare an ISO/IEC 25010-based system evaluation instrument;
- identify adopted questionnaire items from credible references, when appropriate; and
- submit the questionnaire for expert validation.

Directions: Each Capstone Project group shall prepare the following:

Part A. Survey Questionnaire

Develop a survey questionnaire containing:

- Respondent Profile
- Instructions
- Evaluation Items
- Comments and Suggestions

The questionnaire should be aligned with the objectives of the study.

Part B. ISO/IEC 25010 Evaluation Instrument

Construct questionnaire items covering the applicable software quality characteristics.

ISO/IEC 25010 Characteristic	Number of Items
Functional Suitability	
Performance Efficiency	
Compatibility	
Usability	
Reliability	

Security	
Maintainability	
Portability	

Part C. Questionnaire Validation

Submit the developed questionnaire to the assigned validator(s).

Record the recommendations below.

Validator	Recommendation	Action Taken
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Revise the questionnaire based on the recommendations received.

Assessment

The following output shall be evaluated for Module 4.

Output	Assessment Tool
Survey Questionnaire	Evaluation Rubric
ISO/IEC 25010 Evaluation Instrument	Evaluation Rubric
Questionnaire Validation	Validation Checklist

Required Submission

Before the end of Week 4, each Capstone Project group shall submit:

- Survey Questionnaire
- ISO/IEC 25010 Evaluation Instrument
- Adopted Survey Instrument Reference (if applicable)
- Validation Form(s)
- Revised Questionnaire
- Summary of Validators' Recommendations



PRELIMINARY PERIOD

MODULE 5

Interpretation of Data

Week: 5 | Time Allotment: 3 Hours

Module Learning Outcome (MLO 5)

Interpret and present research data using appropriate statistical treatment and organize the results into Chapter IV of the research manuscript.

Target Learning Outcomes (TLO)

At the end of this module, the students should be able to:

- apply appropriate statistical treatment to the collected data;
- interpret research findings based on the computed results;
- prepare Chapter IV of the research manuscript; and
- present the research findings in a clear and organized manner.

Learning Objectives

At the end of this module, the students should be able to:

1. identify the appropriate statistical treatment for the collected data;
2. analyze and interpret the results of the research;
3. prepare tables and figures for Chapter IV;
4. write interpretations that are objective and supported by the data; and
5. present the findings clearly during class discussions or consultations.

Introduction

After collecting data from the respondents, the next phase of the research process is the organization, analysis, and interpretation of the gathered information. The accuracy of the statistical treatment and the clarity of the interpretation directly influence the credibility of the research findings.





This module introduces students to the preparation of Chapter IV of the research manuscript. Students will learn how to organize research data into tables, interpret statistical results, and present findings in a logical and objective manner. Proper presentation of research results enables readers to understand whether the objectives of the study have been achieved.

Statistical Treatment

Statistical treatment refers to the methods used to organize, summarize, and analyze the data gathered from respondents. The selection of an appropriate statistical tool depends on the objectives of the study, the type of data collected, and the research design.

For descriptive studies, commonly used statistical tools include:

- Frequency
- Percentage
- Mean
- Weighted Mean
- Ranking

For studies involving comparisons or relationships, statistical techniques may include:

- t-test
- Analysis of Variance (ANOVA)
- Pearson Correlation
- Chi-Square Test

Researchers should ensure that the statistical treatment selected is appropriate for the objectives of the study and is consistent with the methodology presented in Chapter III.

Data Interpretation

Data interpretation is the process of explaining the meaning of the statistical results obtained from the research.

When interpreting data, researchers should:

- describe the results objectively;
- explain what the numerical values indicate;
- relate the findings to the research objectives;
- avoid unsupported conclusions; and





- present interpretations using clear and concise language.

Interpretations should be based solely on the computed results and should not include personal opinions or assumptions that are not supported by the data.

Writing Chapter IV

Chapter IV presents the findings of the study together with the corresponding analysis and interpretation.

A typical Chapter IV consists of the following sections:

Introduction

A brief statement describing the purpose of the chapter and the organization of the presented data.

Presentation of Results

Each research objective should be addressed separately using appropriate tables or figures.

Interpretation

After every table or figure, provide a narrative interpretation explaining the significance of the results.

Summary of Findings

A concise summary highlighting the major findings of the study.

The presentation of results should follow the sequence of the research objectives to maintain consistency throughout the manuscript.

Presenting Results

Research findings should be presented using properly formatted tables, charts, or graphs whenever appropriate.

When presenting results:

- assign a table number and title;
- arrange data logically;
- use consistent formatting throughout the manuscript;
- label figures and charts clearly; and
- discuss each table immediately after its presentation.

Every table or figure included in Chapter IV should be accompanied by a narrative explanation to help readers understand the significance of the results.

Sample Structure of Chapter IV

A recommended structure for Chapter IV is shown below.

Chapter IV

PRESENTATION, ANALYSIS, AND INTERPRETATION OF DATA

- 4.1 Introduction
- 4.2 Profile of the Respondents
- 4.3 Presentation of Findings Based on Research Objective No. 1
- 4.4 Presentation of Findings Based on Research Objective No. 2
- 4.5 Presentation of Findings Based on Research Objective No. 3
- 4.6 Summary of Findings

This organization allows readers to easily follow the discussion and verify whether each research objective has been addressed.

Tips for Writing Chapter IV

To produce a clear and well-organized Chapter IV:

- Present one research objective at a time.
- Use appropriate statistical treatment.
- Label all tables and figures properly.
- Interpret the results immediately after each table.
- Write objectively and avoid personal opinions.
- Maintain consistency in table numbering and formatting.
- Ensure that every interpretation is supported by the data.

Module Summary

Module 5 focuses on the organization, analysis, and interpretation of research data. Students learn how to apply appropriate statistical treatments, interpret research findings objectively, and prepare Chapter IV of the manuscript. By presenting data using properly formatted tables, figures, and narrative discussions, researchers effectively communicate the results of their study and demonstrate whether the research objectives have been achieved.

References:

Books

- Garcia, L. C., et al. (2016). **Outcomes-Based Practical Guide to Thesis and Capstone Project Writing in Computing**. Mindshapers Co., Inc.
- Howell, D. C. (2008). **Fundamental Statistics for the Behavioral Sciences** (6th ed.).

Institutional References

- Philippine State College of Aeronautics. (2025). *IT411: Aviation Capstone Project 2 course syllabus* (Version 2, Revision 1, 1st Semester, S.Y. 2025–2026). Institute of Computer Studies.
- Philippine State College of Aeronautics. (2013). *PhilSCA student research handbook: AY 2013–2016*.

Standards

- International Organization for Standardization. (2023). ISO/IEC 25010:2023 Systems and software engineering - Systems and software Quality Requirements and Evaluation (SQuaRE) - System and software quality models. ISO.
- International Organization for Standardization & International Electrotechnical Commission. (2023). ISO/IEC 25010:2023 - Systems and software engineering: Systems and software Quality Requirements and Evaluation (SQuaRE) - Product quality model (2nd ed.).

Learning Activity

Activity 5: Data Interpretation Workshop

Objectives

At the end of this activity, each Capstone Project group should be able to:

- organize the collected data using appropriate statistical treatment;
- prepare tables and figures for Chapter IV;
- write interpretations based on the statistical results; and
- present the draft of Chapter IV for adviser feedback.

Directions: Using the data gathered from the respondents, accomplish the following tasks.

Part A. Statistical Treatment

Compute the appropriate statistical treatment based on your approved research methodology.

Prepare the necessary statistical tables for inclusion in Chapter IV.

Part B. Interpretation of Results

For each table prepared, write a narrative interpretation explaining:

- the major findings;
- the significance of the results; and
- how the findings answer the corresponding research objective.

Part C. Drafting Chapter IV

Prepare the initial draft of Chapter IV containing:

- Introduction
- Statistical Tables
- Interpretation of Results
- Summary of Findings

Ensure that the presentation follows the sequence of the research objectives.

Part D. Workshop Presentation

Each group shall present one selected statistical table and explain its interpretation during the class workshop. Faculty members and classmates may provide comments and recommendations to improve the presentation and discussion of the research findings.

Assessment

The following output shall be evaluated for Module 5.

Output	Assessment Tool
Draft Chapter IV	Annotated Manuscript
Statistical Tables	Annotated Manuscript
Interpretation of Results	Annotated Manuscript

Required Submission

Before the end of Week 5, each group shall submit:

- Draft Chapter IV
- Statistical Tables
- Interpretation of Results
- Summary of Findings
- Annotated Manuscript incorporating the comments and recommendations of the adviser or instructor



Preliminary Period

PRELIMINARY ASSESSMENT

Week: 6 | Time Allotment: 3 Hours

Preliminary ASSESSMENT

Overview

The Preliminary Examination serves as the culminating assessment for the Preliminary Period of Capstone Project 2. It provides each Capstone Project group with the opportunity to demonstrate the progress of their research by presenting the initial draft of **Chapter IV** together with the corresponding statistical analysis and interpretation of the collected data.

The examination also evaluates the group's ability to organize research findings, interpret statistical results, and communicate the progress of the study effectively. Faculty members may provide comments and recommendations that will guide the students in improving the succeeding chapters of the research manuscript.

Learning Outcomes Assessed

The Preliminary Examination assesses the students' ability to:

- apply appropriate statistical treatment to the collected data;
- interpret research findings objectively and accurately;
- organize research results into Chapter IV of the manuscript;
- communicate research findings through an oral presentation; and
- respond appropriately to questions and recommendations from the instructor or evaluation panel.

ASSESSMENT Components

The Preliminary Assessment consists of two major components.

Part I

Submission of Chapter IV

Each Capstone Project group shall submit the initial draft of **Chapter IV**.

The manuscript should include:

- Introduction





- Presentation of Results
- Statistical Tables
- Interpretation of Data
- Summary of Findings

The submitted manuscript should follow the approved institutional research format.

Part II

Presentation of Chapter IV

Each group shall conduct an oral presentation of the completed Chapter IV.

The presentation should include the following:

- Brief Introduction
- Research Objectives
- Statistical Treatment Used
- Presentation of Statistical Tables
- Interpretation of Results
- Summary of Findings
- Questions and Answers

Each group member is expected to actively participate during the presentation.

General Guidelines

Before the examination, each group should ensure that:

- Chapter IV has been proofread.
- Statistical computations have been verified.
- Tables and figures are properly labeled.
- Presentation slides are complete.
- All members are familiar with the research findings.
- Supporting files are available before the scheduled presentation.

Required Submission

Each group shall submit the following:



- Draft of Chapter IV
- Statistical Tables
- Presentation Slides
- Soft Copy of the Manuscript (if required)
- Supporting Statistical Computations (if required)

Assessment Criteria

The Preliminary Examination shall evaluate the following:

Component	Assessment Tool
Chapter IV Manuscript	Manuscript Evaluation Rubric
Oral Presentation	Presentation Rubric
Statistical Interpretation	Faculty Evaluation
Response during Question and Answer	Oral Presentation Rubric

Expected Output

At the end of the Preliminary Examination, each Capstone Project group should have successfully:

- submitted the initial draft of Chapter IV;
- presented the research findings using appropriate statistical treatment;
- received recommendations from the instructor or evaluation panel; and
- identified the necessary revisions for the succeeding phase of the research.